Week 3 – Data Analysis, Modelling Skills and Lab Safety (Lean Notes)

Lectures covered: L14 (data analysis), L15 (R demo), L16 (modelling skills), L17 (R modelling demo), L18 (safety in the laboratory). ⭐ = tested in the Week 3 assignment.

1. Concept cards (in the order taught)

1. What is data analysis

- Extracting **useful, relevant, meaningful information from observations**. Pushing data into software and showing graphs is only a tiny part. Every research (even theoretical) ends up needing data.
- Purposes: **parameter estimation** (any unknown: slope, mean, variance, median), **model development** (prediction = forecasting), **feature extraction** (peaks in an ECG), **classification**, **hypothesis testing** (claim → check evidence in data), **fault detection / process monitoring** (compare with normal pattern).
- Flow: process → sensors (a survey taker is a "human sensor") → data → analysis (visualise, estimate, model) → use (monitoring, control, optimisation). **Domain knowledge** is used at every step.

2. Types of analysis

Exploratory (no claim upfront, plots and summary statistics) vs confirmatory (test a postulate) · qualitative (trends, oscillations) vs quantitative (numbers) · descriptive (first step) vs inferential (hypothesis tests) · predictive (where is it heading) vs prescriptive (why did it happen and how to prevent it). Start with exploratory/descriptive: **"make friends with the data"**. The end use decides the type.

3. Deterministic vs stochastic data ⭐

- The words describe the **data generating process (DGP)** = physical process + experiment + **sensor**, not the data alone.
- **Deterministic:** a mathematical function explains every data point exactly (e.g. a sine wave plus a rectangular pulse; a ball dropped in still air). **Stochastic/random:** not exactly predictable, outcome is one of many possibilities, described by probability (rain tomorrow, sensor noise, stock index).
- In reality nothing is perfectly deterministic (sensors are never perfect); it is an idealisation. **Mixed** is common: reactor physics well known (deterministic) + sensor noise (stochastic).
- Calling data stochastic is a formal way of saying *"I do not understand the process enough"*; also state which kind (stationary, non-stationary, periodic). There is no formula to decide; go by what you know. The choice changes the theory, tools and interpretation (e.g. what "mean" and "periodic" mean).
- **Measured signals and estimates from measurements are random.** Random signals can be negative; "random" is not "occurs randomly" (it follows probability laws).

4. Estimates are random: key terms ⭐

- Any estimate computed from random data is itself **random**: it is one of many values you could have got. The analyst must report its **error**, and a good method has error that **shrinks to zero as n → ∞**.
- **Population** = all possibilities; **sample** = the one data record you have. **Truth** (population value) vs **estimate**.
- **Statistic** = any mathematical function of the observations (sample mean, sample variance). **Estimator** = such a function designed to estimate something. An estimate **is a function of the observations**.

- **Bias** = systematic error: the *average of the estimate over many data records* ≠ truth. Measures **accuracy**. **Variability** = spread of estimates across data records. Measures **precision** ("high-precision instrument" = repeated readings vary little).
- **Standard deviation of the estimate (standard error)** quantifies the error in an estimate. Mode, median, skewness do not.
- **Efficient** estimator = **lowest standard error among all estimators**. **Robust** = not hurt by outliers. Sample mean is most efficient for Gaussian, outlier-free data; sample median is robust when outliers exist. Least squares is not always the lowest-error method.

5. Procedure of data analysis

- Iterative, not strictly sequential. Success = **data quality + systematic method**. If data are too poor, return them ("do not expect miracles"). Data from a **faulty sensor are unreliable** because they describe the sensor, not the process.
- **Preliminary steps: visualise** first, **quality-check**, **pre-process**. **Anscombe's data sets:** four sets with the same mean, variance, correlation and regression line but totally different plots (curve, outlier). Handle **missing values and outliers**, sync time stamps and sampling rates, filter noise, keep part of the data for **cross-validation**, watch quantised/compressed historian data. **80/20 rule:** up to 80% of time goes to pre-processing.
- Ask first: type of analysis? source? how acquired (sensor, sampling rate)? how informative? **what assumptions** about the DGP?
- **Core:** choose the **domain** (time or frequency, wavelets), **reduce dimensions** if data are huge (PCA, non-negative matrix factorisation), choose the right tool (mean vs median). No unique answer.
- **Report:** test whether each parameter is significant (null: parameter = 0), give **confidence regions and errors** (never report estimates without errors), cross-validate, pick among models by **AIC/BIC** and end use (simple model preferred). Data analysis is **science and art**.

6. R demo (L15)

- **R** (with RStudio) = free, open-source statistics software. Built-in data sets; `plot`, `summary`, `cor`, `t.test`.
- Anscombe pairs: same means and correlation, very different pictures.
- Beaver temperatures: treated as stochastic (a 113th-degree polynomial would fit but predict badly). **t-test** compares two means: null "means are equal"; **low p-value ⇒ reject the null** ("if p is low, the null must go"); a 95% confidence interval for the difference that excludes 0 also rejects. Check assumptions (the t-test expects Gaussian data).
- CO_2 series: trend + oscillation + leftover noise = **mixed** deterministic and stochastic.

7. What is a model, and why

- A **model** = mathematical or descriptive abstraction of a process. Uses: **prediction**, **classification**, **fault detection** (alarm when measurement differs from model prediction), **simulation** (needs high fidelity), **control** (approximate model is fine), design/optimisation. The end use decides the type.
- Model inputs are not the same as process inputs: a dynamic model uses the **present and past** inputs.

8. First-principles vs empirical vs grey box

Approach	Idea	Pros	Cons
First principles (white box, mechanistic)	conservation of mass/energy/momentum plus laws	causal, physical meaning, works over wide conditions	hard ODE/PDE, no freedom in structure
Empirical (black box, data driven)	fit input-output data	flexible, simple, quick	only as good as data, doubtful extrapolation, needs estimation algorithms
Grey box	empirical + prior knowledge	best working solution	–

Random (stochastic) processes need empirical **time series models**. Car test-drive = building an empirical model in your head.

9. Linearisation and the tank example ⭐

- Liquid tank: at steady state outflow ∝ √(level) (first principles); mass balance gives a **non-linear first-order ODE**; **Taylor series** about a nominal operating point (first-order terms kept) gives a **linear model in deviation variables**. Empirical grey-box version: a **discrete-time first-order difference equation** whose parameters a and b are estimated (ε = error); black box: choose structure yourself.

$$f(x) \approx f(x_0) + f'(x_0)\,(x - x_0)$$

Terms: x_0 = nominal point; $f'(x_0)$ = slope there; $x - x_0$ = deviation from nominal.

What it is for: replaces a curve by its tangent line near x_0, so linear tools apply; the line changes if x_0 changes.

Tiny number: f(x) = √x at $x_0 = 4$: $\sqrt{x} \approx 2 + (x - 4)/4$. At x = 5 gives 2.25 (true 2.236).

10. Systematic modelling procedure

Three stages: **data acquisition** → **model development** (pre-process → optional **non-parametric analysis** to guess structure → **parametric estimation** by optimisation) → **model assessment** (fresh data, errors in parameters, **over-fitting**). Even first-principles models must be validated. Iterate; use domain knowledge at every stage.

11. Two model classes

- **Time series models:** stochastic, causes unknown or known with error (atmospheric temperature from its own history).
- **Input-output (causal) models:** explain one variable by **regressors** (explanatory variables). Regressor known exactly if you set it (coolant flow); known with error if measured (temperature predicting humidity). Real models are a **mix** (sensor noise needs a time-series part).

12. Excitation: data must carry enough information

- Steady state $y = b_0 + b_1u + b_2u^2$ has 3 unknowns, so you need **three different steady states**; the input matrix must be **full rank** (non-singular).
- Dynamic system with 3 parameters: a **single-frequency sine** input makes the matrix singular (only 2 parameters identifiable); use **at least two frequencies**.

13. Noise and signal-to-noise ratio ⭐

- **SNR** = power (squared amplitude) of the true signal ÷ power of the noise. Noise hurts prediction accuracy and parameter estimates. **Higher SNR ⇒ smaller error.**

$$\sigma_{\hat{\theta}} \propto \frac{1}{\sqrt{\text{SNR}}} \qquad\qquad \sigma_{\hat{\theta}} \propto \frac{1}{\sqrt{n}}$$

Terms: σ of $\hat{\theta}$ = standard error of the parameter estimate; SNR = signal-to-noise ratio; n = number of data points.

What it is for: tells how much cleaner data or more data shrink the error (to halve the error you need 4 times the SNR or 4 times the data).

Tiny number: SNR 9 vs 100: errors ∝ 1/3 and 1/10, ratio **10 : 3**. Lecture demo: SNR 100 vs 10 gave standard errors 0.036 vs 0.114 (about 3.2×, which is √10).

14. Over-fitting

- Over-fitting = treating **local chance fluctuations (noise) as the real trend**. A 99th-degree polynomial passes through 100 noisy points yet predicts new data badly (like a student memorising assignment numbers).

- **Symptoms:** (1) **large standard errors / insignificant coefficients** (coefficient significance read from p-value or stars), (2) **poor prediction on fresh (test) data**. Under-fitting (too low an order) is also bad.
- Cures: **train/test split** (e.g. first 150 points to fit, last 51 to test), plot residual sum of squares against model order and stop when gains become tiny, keep the model **as simple as possible**. "Data is the food, parameters are the guests": too many parameters and some go hungry (large errors).

15. Reflection checklist for any model

Choose the simplest adequate model type; model deterministic **and** stochastic parts; the experiment decides model quality (excitation, low noise; see Design of Experiments, Week 7); pick an **efficient** estimation method (least squares, maximum likelihood, Bayesian); error falls as 1/√n; choose time or frequency domain.

16. R modelling demo (L17)

- `lm` = linear model by least squares. Coefficient table: **estimate, standard error, t value, p value, stars**. Read an estimate **relative to its standard error** (a tiny-looking value can be significant, units matter).
- Fit trend, look at **residuals**, find the next pattern. CO_2: linear trend → still quadratic trend left → oscillation left.
- **Fourier transform / periodogram** (power spectral density): a **peak at a frequency = oscillation at that frequency** (used to detect periodic components); a smaller peak at a multiple is a **harmonic** (hint of non-linearity).
- Polynomial demo: cubic fits well; 4th and 5th order give insignificant coefficients and bad test predictions.

17. Safety in the laboratory ⭐

- Industry jobs are well defined; a **research lab keeps pushing boundaries**, so you must work out for yourself what needs care. Safety protects **people, equipment and the experiment**, and is a **culture**, not a few rules.
- Common hazards: **chemicals** (acids, bases, powders, biological samples), **electricity**, **water**, **pressurised gas bottles**, **heat sources** (hot plate, oven, furnace up to 1000–1500 °C).
- **Protective gear: lab coat**; **gloves** (nitrile for general use; check the glove suits the solvent; thermal gloves for hot items; acid-resistant gloves); **safety glasses** are **not ordinary spectacles** (spectacles leave sides open and can shatter); safety glasses go over prescription glasses and are needed even for hammering a nail; goggles/face shield for glare; **dust mask** (surgical type may be inadequate; filter respirator for fine dust).
- **Gas bottles:** the neck with the regulator is delicate; if it falls it can break and release gas. **Chain the bottle to the wall at about 2/3 to 3/4 of its height** (too low is useless); use the same restraint on trolleys; never leave a regulated bottle unsecured.

2. Top traps (most likely first)

1. **Deterministic vs random:** sine plus pulse = deterministic; measured or estimated quantities = random; real processes are often **mixed**; random signals can be negative.
2. An estimate is a **function of the data** (so it is random); it is **not** guaranteed close to the truth, averaging a million records gives the truth only if unbiased, and least squares is not always best.
3. Error ∝ **1/√SNR** (not 1/SNR); noisier data = larger error, so SNR 9 vs 100 gives 10 : 3.
4. **Standard deviation (standard error)** measures estimate error, not mode, median or skewness.
5. **Linearisation** = first-order Taylor expansion about a nominal point (not "dropping terms"); the model changes with the point; non-linear models can output zero for non-zero input (sin x); linear models do not automatically have smaller errors.
6. **Fourier transform** detects periodic components (not Hilbert or z-transform).
7. **Efficient** = lowest standard error among estimators (not fastest, not fewest samples, not outlier-proof).
8. Data from a **faulty sensor cannot be trusted** or corrected by assuming the error.
9. A perfect fit to training data is **over-fitting**, not success; check parameter errors and test data.

10. Safety glasses ≠ spectacles; gas bottle chain at 2/3 to 3/4 height; heat, chemicals, gas cylinders, electricity and water are all hazards.
11. Lecture slips: “SNR down by 10 gives about 3.6 times the error” is really **√10 ≈ 3.16** (the numbers shown, 0.114/0.036, are about 3.2). Anscombe’s four sets: same statistics, different pictures.

3. Assignment questions with solutions

Q1. Safety is the most important discipline in a lab. Common sources that can cause damage are (select all that apply) *(Q1)* - A. Sources of heat B. Chemicals (acids, bases etc.) C. Pressurised gas cylinders D. Electricity and water source

Solution (card 17): The lecture lists all four as standard hazards. **Answer: A, B, C and D.**

Q2. Which statistic quantifies errors in parameter estimates? *(Q2)* - A. Mode B. Skewness C. Standard deviation D. Median

Solution (card 4): Error of an estimate = its spread (standard error). Mode and median give centre, skewness gives asymmetry. **Answer: C.**

Q3. A temperature sensor at the bottom of a cooling-tower tank, measuring outlet water temperature, malfunctioned for some time and gave higher values. The conclusion is: *(Q3)* - A. Different values increase the scope of analysis, so they are useful - B. The data cannot be relied on, as they come from a malfunctioning sensor - C. The data can be used by assuming the error and subtracting it - D. None of these

Solution (card 5): Data quality limits any analysis; the size of a sensor fault is unknown, so it cannot be subtracted. **Answer: B.**

Q4. Which can be treated as a deterministic signal? *(Q4)* - A. Measurements of atmospheric temperature B. A sine wave corrupted by a rectangular pulse - C. Estimates of signals from measurements D. None of the above

Solution (card 3): A and C are random (measurements and estimates carry randomness). A sine wave plus a pulse is a fixed mathematical function. **Answer: B.**

Q5. Identify the INCORRECT statement(s) about non-linear models (select all that apply) *(Q5)* - A. Linearisation involves mere discarding of the non-linear terms - B. The Taylor-series linearised model changes with the nominal point - C. A non-linear model can never yield a zero output for a non-zero input - D. Estimates of linear models have smaller errors than those of non-linear models

Solution (card 9): A is false (it is a first-order Taylor expansion). B is true (slope depends on the point). C is false ($\sin x = 0$ at $x = \pi$). D is false (error depends on data and fit). **Answer: A, C and D.**

Q6. Which transform detects periodic (oscillatory) components in a signal? *(Q6)* - A. Hilbert B. z-transform C. Fourier D. All of the above

Solution (card 16): A peak in the Fourier power spectrum shows an oscillation at that frequency. **Answer: C.**

Q7. The ratio of errors in parameter estimates of a model from two data sets with SNR 9 and 100 is *(Q7)* - A. 3 : 10 B. 5 : 1 C. 10 : 3 D. 10 : 0.9

Solution (card 13): Error $\propto 1/\sqrt{\text{SNR}}$. For SNR 9: 1/3. For SNR 100: 1/10. Ratio (1/3) : (1/10) = 10 : 3 (noisier set has the larger error). **Answer: C.**

Q8. Which is true about a parameter estimate? *(Q8)* - A. It is in general close to the true value with high probability - B. Averaged over a million records under the same conditions, it equals the true value - C. Least squares always gives the lowest-error estimate - D. It is a mathematical function of the observations

Solution (card 4): An estimate is a formula applied to data, so D. A, B and C promise guarantees (closeness, no bias, best error) that are not true in general. **Answer: D.**

Q9. Which statements are true? (select all that apply) *(Q9)* - A. Deterministic phenomena produce outcomes that can be predicted accurately - B. Random processes occur randomly - C. A process can produce a mix of deterministic and stochastic effects - D. Random signals can never take negative values

Solution (card 3): A and C are the lecture's points. B says nothing (random processes follow probability laws). D is false (noise goes negative). **Answer: A and C.**

Q10. An estimator is most efficient when *(Q10)* - A. Parameters are estimated in the least time - B. It gives accurate estimates with the smallest number of observations - C. Its standard error is the lowest compared with all other estimators - D. It estimates accurately even in presence of outliers

Solution (card 4): Efficiency compares standard errors; D describes robustness. **Answer: C.**

Answer key: Q1 A,B,C,D · Q2 C · Q3 B · Q4 B · Q5 A,C,D · Q6 C · Q7 C · Q8 D · Q9 A,C · Q10 C